Electric Vehicle Kit



Category: **Automotive** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Electric Vehicle Kit (two-wheeler); Electric Vehicle Training Kit

What it is used for

An electric vehicle training kit demonstrates the main electrical, electronic, and mechanical systems used in an electric two-wheeler or EV platform. Correct setup, controlled testing, isolation procedures, and supervision help learners understand EV operation while reducing electric shock, short-circuit, and battery hazards.

Main parts

- Traction battery or low-voltage training power source supplies electrical energy.
- Motor and drive assembly demonstrate propulsion.
- Motor controller or inverter regulates motor operation.
- Throttle or command input provides a driving request.
- DC-DC converter demonstrates voltage conversion where included.
- Wiring harness and connectors link system components.
- Display or instrument panel shows operating information.
- Brake switches and sensors demonstrate safety inputs where included.
- Charging port or charger interface demonstrates charging functions where supplied.
- Emergency stop and isolation device disconnect power where fitted.

Before you start

1. Only trained instructors and authorised learners should use the kit under supervision.
2. Read the kit manual and identify whether the setup uses low-voltage training power or a traction-voltage battery.
3. Inspect the frame, battery, motor, controller, connectors, cables, charger, guards, and emergency stop for damage.
4. Confirm the kit's voltage, current, and power ratings and verify that all components are compatible.
5. Use the specified charger and check battery condition; do not use a swollen, leaking, overheated, or damaged battery.
6. Establish a clear work area and identify the isolation point, emergency stop, and required PPE.
7. For traction-voltage systems, only qualified and authorised personnel may perform high-voltage work using the prescribed isolation and verification procedure.

How to use

1. Place the training kit on a stable surface and secure any moving assembly according to the manual.

2. Confirm the emergency stop and isolation controls are accessible and functional before energising the kit.
3. With power isolated, inspect wiring diagrams and identify the battery, controller, motor, sensors, and communication connections.
4. Connect only the approved low-voltage supply or battery using the specified polarity and sequence.
5. Check connector locking, cable routing, protective covers, and any required fuses before starting.
6. Energise the kit according to the instructor's procedure and observe the display for warnings or fault indicators.
7. Demonstrate throttle input, motor response, sensor readings, and control functions at the approved training level.
8. Keep the motor output and moving components within the kit's specified operating limits during tests.
9. Use diagnostic software or measurement instruments only according to the training instructions and equipment ratings.
10. If abnormal heating, smoke, unusual noise, unexpected motor movement, or a fault occurs, activate the emergency stop and isolate power when safe.
11. After testing, return controls to the safe state, switch off the supply, and follow the specified shutdown sequence.
12. Disconnect and store accessories as instructed, record test results, and report faults before the next session.

⚠ Safety

- Never bypass the emergency stop, interlocks, fuses, or other protective devices.
- Do not touch exposed conductors or connect or disconnect power cables while energised unless the approved procedure explicitly permits it.
- Never short-circuit a battery or use an unapproved charger or power supply.
- Keep hands, clothing, and tools away from rotating motors, wheels, belts, and drive components.
- Do not work on traction-voltage circuits without required qualifications, isolation, voltage verification, and appropriate PPE.
- Stop use immediately if the battery is swollen, leaking, smoking, unusually hot, or physically damaged; follow the site's battery emergency procedure.

Care & storage

- Inspect cables, connectors, battery enclosure, motor mountings, and protective devices before each training session.
- Keep the kit clean, dry, and free from conductive debris or loose metal objects.
- Charge and store batteries according to the manufacturer's temperature and state-of-charge guidance.
- Update training software and maintain measurement equipment according to approved procedures.
- Remove damaged equipment from service and arrange authorised inspection or repair.

Common mistakes to avoid

- Connecting components without checking the wiring diagram, polarity, and voltage ratings.
- Using a battery or charger that is incompatible with the kit.
- Testing motor functions with people or objects inside the moving-component zone.
- Attempting high-voltage repairs without qualification or proper isolation.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.